A Synopsis Report
on

Impact of Database Management Systems on Organizational Performance in an Industry

Submitted in partial fulfillment
for the award of the degree of

Bachelor of Technology
in
Computer Science Engineering (Artificial Intelligence & Data Science)

Submitted By

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Under the Supervision of
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Project Credentials

Project title	Impact of Database Management Systems on Organizational Performance in an Industry
Project Domain/Area	Database Management Systems, Industrial Information Systems
Group members	1. kartik 2822530 2. Sushant 2822532 3. Shubham Pangal 2822512
Group Id *(to be allotted by project coordinator)*	G19
Supervisor's Name	Dr. B.K Verma
Supervisor's Designation	HOD CSE-AIDS

Supervisor's Consent

The synopsis of final year project work titled…… by the students' group id.…G19…… has been written with my consent and every section of this synopsis report is reflecting the work to be carried out by the group.	*(Signature of supervisor with date)*

Department Project Evaluation Committee (DPEC) Remarks

The project is ..... by DPEC. The group is advised to submit progress of the project work in progress presentation-1 to be held on…….

OR

The project is ..... by DPEC. The group is advised to submit the synopsis report again after making changes as suggested by DPEC on …….....

Name & Signature of DPEC member (s) with date

Impact of Database Management Systems on Organizational Performance in an Industry

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Recommendation with Engineering Program Outcomes and Program/Department Specific Outcomes

Project Title	Engineering Program Outcomes (POs)	Department Specific Outcomes (DSOs)
Impact of Database Management Systems on Organizational Performance in an Industry	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO11	PSO1, PSO2

Project Categorization

• Research-based:

The project involves a systematic study of how a custom-built relational DBMS (FactoryFlow) impacts productivity, defect rates, and decision-making in manufacturing organizations. Data analysis is performed on real production datasets to quantify performance improvements.

• Application-based:

FactoryFlow is a fully functional industrial DBMS web application providing employee management, production tracking, AI-driven analytics, executive dashboards, and decision-support tools — directly deployed on cloud infrastructure.

Project Title	Project Categorization
Impact of Database Management Systems on Organizational Performance in an Industry	Application-based (primary) + Research-based

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Project Map with SDG Goals

Sustainable Development Goals (SDGs):

- **SDG 8 – Decent Work and Economic Growth:** FactoryFlow improves worker productivity tracking and optimizes resource allocation, directly contributing to economic growth in manufacturing enterprises.
- **SDG 9 – Industry, Innovation and Infrastructure:** The project demonstrates how modern DBMS and AI technologies form the backbone of industrial information infrastructure and drive manufacturing innovation.

- **SDG 12 – Responsible Consumption and Production:** By minimizing defects, identifying inefficiencies, and reducing data redundancy, FactoryFlow promotes responsible and sustainable production practices.

• Justification

In modern manufacturing industries, massive volumes of operational data are generated daily — from production floor records and employee attendance to financial transactions and quality metrics. Traditional spreadsheet-based or paper-based systems are error-prone, fragmented, and lack the real-time visibility that management needs to make informed decisions.

FactoryFlow addresses this gap by implementing a cloud-hosted relational DBMS (PostgreSQL) accessible via a responsive web interface. The system provides a single source of truth for all organizational data. AI-powered modules analyze production trends, detect anomalies, and recommend strategic interventions — directly validating the thesis that a well-designed DBMS significantly enhances organizational performance.

The project is justified by the measurable outcomes it delivers: real-time efficiency scores, defect rate tracking, revenue and profit analytics, workforce productivity metrics, and AI-generated strategic recommendations — all derived from the underlying PostgreSQL database.

Introduction

Database Management Systems are the foundation of any data-driven industrial organization. As manufacturing industries grow in scale and complexity, the ability to capture, store, query, and analyze operational data in real time becomes a critical competitive advantage.

The project titled **"Impact of Database Management Systems on Organizational Performance in an Industry"** is realized through **FactoryFlow** — a full-stack industrial DBMS web application. FactoryFlow manages the complete operational lifecycle of a manufacturing organization: employee records, production batches, departmental structure, role hierarchies, financial tracking, and AI-powered performance analytics.

The system is built on Next.js 15, PostgreSQL (Neon Cloud), and Prisma ORM, with an AI layer powered by the Groq API (LLaMA 3.3 70B). It provides an executive dashboard, AI Command Center (4 AI tools), real-time KPI monitoring, anomaly detection, production forecasting, audit logs, and exportable PDF reports — demonstrating the full potential of modern DBMS in industrial settings.

Objectives and Scope

Objectives:

- To design and develop a full-stack Industrial DBMS web application (FactoryFlow) managing employees, production, departments, and roles.
- To implement complete employee lifecycle management including hiring, role promotion, status tracking, and department assignment.
- To build a production batch tracking system with real-time KPI analytics (units, defects, efficiency, revenue, profit).
- To integrate an AI Command Center with four AI engines: Predictive Maintenance, Resource Optimization, Quality Control AI, and Performance Benchmark.
- To provide an executive decision-support dashboard with AI-based production forecasting and anomaly detection.
- To quantify the measurable impact of DBMS implementation on industrial organizational performance.

Scope:

The scope encompasses a complete cloud-deployed web-based DBMS for a manufacturing organization. It covers employee management (CRUD, promotions, audit logs), production record management, financial analytics (revenue, profit, cost per unit), AI-driven anomaly detection over the last 60 days of production, predictive production forecasting, department and role hierarchy management, DBMS benchmark comparison, and an intelligent floating AI chat assistant capable of performing database operations via natural language.

Methodology

The project follows an Agile development methodology with iterative sprints. The database schema is designed using Prisma ORM with entity relationships covering Employee, Department, Role, Product, Production, PromotionHistory, AuditLog, and Dataset tables. All business logic is exposed through REST API routes (Next.js). The frontend consumes these APIs using React hooks with real-time event-driven cache invalidation. AI analytics are implemented via structured prompts to the Groq API, with JSON-parsed responses rendered as premium UI dashboards. Testing is performed against live PostgreSQL data on Neon Cloud.

Hardware & Software Requirements

Hardware:

- Minimum 8 GB RAM (16 GB recommended)
- Intel i5 / Ryzen 5 processor or higher

• Personal computer with stable internet connection

• Cloud server — Vercel (frontend/API) + Neon PostgreSQL (database)

Software:

Category	**Technology**
Framework	Next.js 15 (React 19, App Router)
Database	PostgreSQL 16 — Neon Cloud (serverless)
ORM	Prisma ORM 6
AI / LLM	Groq API — LLaMA 3.3 70B Versatile (streaming)
Styling	Tailwind CSS, Framer Motion, CSS Variables
Charts	Recharts, custom SVG gauges
Auth / Security	Role-based access, audit logging, Prisma soft-delete
Dev Tools	VS Code, Git, GitHub, Postman
OS	Windows 11 / Linux (Ubuntu)

Limitations

- AI analytics accuracy improves with larger production datasets; limited data may reduce prediction confidence.
- Real-time multi-factory synchronization across geographically distributed locations is not yet implemented.
- The AI chat and analytics modules depend on the availability of the third-party Groq API service.
- The current system is optimized for single-organization deployment and does not yet support multi-tenant SaaS architecture.
- PDF export reports are generated server-side and may be slow for very large datasets.

Future Scope

- Multi-factory support with a centralized DBMS and cross-site analytics dashboard.
- IoT sensor integration for automated real-time production data ingestion from factory floor devices.
- Mobile application (React Native / PWA) for on-floor data entry and instant KPI monitoring.
- Advanced ML models (TensorFlow / PyTorch) for predictive quality control and maintenance scheduling.
- ERP system integration with SAP, Oracle, or Microsoft Dynamics for enterprise-level data flow.
- Role-based multi-tenant SaaS deployment supporting multiple industrial clients on shared infrastructure.
- Natural language SQL query interface allowing managers to query the database in plain English.

Conclusion

FactoryFlow successfully demonstrates the transformative impact of a well-designed Database Management System on organizational performance in a manufacturing industry. By consolidating employee records, production data, financial metrics, and quality indicators into a single PostgreSQL DBMS, the system eliminates data silos, reduces redundancy, and enables data-driven decision-making at every level of the organization.

The integration of AI-powered analytics — including production forecasting, anomaly detection, resource optimization, and predictive maintenance — further amplifies the value of the underlying DBMS by converting raw operational data into actionable intelligence. Real-time KPI dashboards, audit trails, and exportable reports empower management to respond swiftly to performance changes and strategic opportunities.

This project validates the thesis that modern DBMS implementations, when combined with intelligent analytics, directly and measurably enhance industrial organizational performance across

efficiency, quality, workforce management, and financial dimensions.

References

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